

Glucagon: Drug information - UpToDate

Glucagon: Drug information - UpToDate

Official reprint from UpToDate®

www.uptodate.com © 2026 UpToDate, Inc. and/or its affiliates. All Rights Reserved.

Glucagon: Drug information

2026© UpToDate, Inc. and its affiliates and/or licensors. All Rights Reserved.

Contributor Disclosures

For additional information see "Glucagon: Patient drug information" and "Glucagon: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Baqsimi One Pack;
- Baqsimi Two Pack;
- GlucaGen Diagnostic [DSC];
- GlucaGen HypoKit [DSC];
- Gvoke HypoPen 1-Pack;
- Gvoke HypoPen 2-Pack;
- Gvoke Kit;
- Gvoke PFS;
- Gvoke VialDx [DSC]

Brand Names: Canada

- Baqsimi

Pharmacologic Category

- Antidote;
- Antidote, Hypoglycemia;
- Diagnostic Agent

Dosing: Adult

Dosage guidance:

Clinical considerations: Consider pretreatment with antiemetic when using rapid administration for anaphylaxis, beta-blocker overdose, or calcium channel blocker overdose, particularly in patients with severe drowsiness or who are obtunded (Ref).

Anaphylaxis in patients on beta-blocker therapy

Anaphylaxis (refractory to epinephrine) in patients on beta-blocker therapy (off-label use): IV: Initial: 1 to 5 mg bolus; followed by an infusion of 5 to 15 **mcg/minute**; titrate the infusion rate to achieve an adequate clinical response (Ref).

Beta-blocker overdose

Beta-blocker overdose (off-label use): IV: 2 to 10 mg bolus; if no clinical response, may repeat bolus dose after 10 to 15 minutes; if clinical response with bolus, start continuous infusion at 2 to 5 mg/hour (range: 1 to 15 mg/hour); titrate infusion rate to achieve adequate hemodynamic response (Ref).

Calcium channel blocker overdose

Calcium channel blocker overdose (off-label use): IV: 2 to 10 mg bolus; if no clinical response, may repeat bolus dose twice at 2 minute intervals; if clinical response with bolus, start continuous infusion at 2 to 5 mg/hour (range: 1 to 15 mg/hour); titrate infusion rate to achieve adequate hemodynamic response (Ref).

Diagnostic aid, radiologic examinations

Diagnostic aid, radiologic examinations:

Relaxation of colon:

IM: 1 to 2 mg.

IV: 0.5 to 0.75 mg.

Relaxation of stomach, duodenal bulb, duodenum, and small bowel:

IM: 1 mg.

IV: 0.2 to 0.5 mg.

Growth hormone deficiency, alternative diagnostic test

Growth hormone deficiency, alternative diagnostic test (off-label use): Note: Correct other pituitary hormone deficiencies prior to administration (Ref).

IM:

Weight ≤90 kg: 1 mg as a single dose.

Weight >90 kg: 1.5 mg as a single dose.

Hypoglycemia

Hypoglycemia:

Injection:

GlucaGen HypoKit or Glucagon Emergency: **IM, IV, SUBQ:** 1 mg; may repeat in 15 minutes as needed.

Gvoke HypoPen, Gvoke PFS, or Gvoke Kit: **SUBQ:** 1 mg; may repeat in 15 minutes as needed using a new device.

Intranasal: 3 mg (one actuation) into a single nostril; if no response, may repeat in 15 minutes using a new intranasal device.

Dosing: Kidney Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling. However, data suggest need for dosage adjustment unlikely (Ref).

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Glucagon: Pediatric drug information")

Hypoglycemia, severe; treatment

Hypoglycemia, severe; treatment: Note: Glucagon should be prescribed for all patients with type 1 or type 2 diabetes who are taking insulin or are at high risk for hypoglycemia; caregivers, school personnel, and family members of these patients should be trained on when and how to administer glucagon (Ref). Patients with inadequate glycogen stores (eg, starvation, adrenal insufficiency, chronic hypoglycemia) may not respond to glucagon and should be treated with glucose (Ref).

Parenteral:

Weight-based dosing: Infants, Children, and Adolescents: IM, IV, SUBQ: 0.02 to 0.03 mg/kg/dose; may repeat every 15 minutes if needed for clinical effect for up to a total of 3 doses; maximum dose: 1 mg/dose (Ref).

Product-specific dosing:

Glucagon: Infants, Children, and Adolescents:

<20 kg: IM, IV, SUBQ: 0.5 mg; if no response in 15 minutes, may repeat dose.

≥20 kg: IM, IV, SUBQ: 1 mg; if no response in 15 minutes, may repeat dose.

GlucaGen: Infants, Children, and Adolescents:

Weight-directed dosing:

<25 kg: IM, IV, SUBQ: 0.5 mg; may repeat once if needed.

≥25 kg: IM, IV, SUBQ: 1 mg; may repeat once if needed.

Age-directed dosing: **Note:** Use when weight is unknown.

Infants and Children <6 years: IM, IV, SUBQ: 0.5 mg; may repeat once if needed.

Children ≥6 years and Adolescents: IM, IV, SUBQ: 1 mg; may repeat once if needed.

Gvoke:

Children 2 to <12 years:

<45 kg: SUBQ: 0.5 mg; if no response after 15 minutes, may administer an additional 0.5 mg dose.

≥45 kg: SUBQ: 1 mg; if no response after 15 minutes, may administer an additional 1 mg dose.

Children ≥12 years and Adolescents: SUBQ: 1 mg; if no response after 15 minutes, may administer an additional 1 mg dose.

Intranasal (Baqsimi):

Children and Adolescents: Intranasal: 3 mg (1 actuation) intranasally into a single nostril; if no response after 15 minutes, may repeat dose using a new device. **Note:** Inhalation not necessary.

Hypoglycemia, mild or impending hypoglycemia during illness

Hypoglycemia, mild (<70 mg/dL) or impending hypoglycemia during illness (mini-dose regimen): Limited data available (Ref): **Note:** These doses are lower than those used for treatment of severe hypoglycemia and have been shown to prevent hypoglycemia for several hours during mild hypoglycemia or impending hypoglycemia-associated illness (eg, gastroenteritis, nausea/vomiting) and/or poor oral carbohydrate intake (Ref).

Initial dose:

Infants and Children ≤2 years: SUBQ: 0.02 mg.

Children >2 years and Adolescents ≤15 years: SUBQ: 0.01 mg per year of age.

Adolescents >15 years: SUBQ: 0.15 mg.

Subsequent dose: SUBQ: If initial dose fails to increase glucose after 30 minutes, may give an additional dose that is twice the initial dose.

Anaphylaxis in patients on beta-blocker therapy

Anaphylaxis (refractory to epinephrine) in patients on beta-blocker therapy: Limited data available (Ref):

Infants, Children, and Adolescents: IV:

Initial: 0.02 to 0.03 mg/kg slow IV; maximum dose: 1 mg/dose.

Continuous infusion: Follow initial dose with a continuous infusion of 5 to 15 **mcg/minute**; titrate to effect.

Beta-blocker or calcium channel blocker toxicity/overdose

Beta-blocker or calcium channel blocker toxicity/overdose: Limited data available:

Infants and Children:

Loading dose: IV: 0.05 mg/kg as a single dose (Ref); if no response, may repeat dose (Ref).

Continuous IV infusion: 0.05 to 0.1 mg/kg/**hour**; titrate to effect (Ref); usual adolescent dose: 1 to 5 mg/**hour** (Ref); some experts recommend initiating the infusion at the "response dose" (cumulative loading dose) per hour (Ref).

Adolescents:

Loading dose: IV: 5 to 10 mg (Ref).

Continuous IV infusion: 1 to 5 mg/**hour** (Ref); some experts recommend initiating the infusion at the "response dose" (cumulative loading dose) per hour (Ref).

Esophageal food impaction, esophageal relaxation

Esophageal food impaction, esophageal relaxation: Limited data available, efficacy results variable (Ref):

Children and Adolescents: IV: 1 mg once prior to endoscopic intervention; may repeat once within 15 to 30 minutes if needed and no adverse effects observed with first dose. **Note:** Glucagon administration should not delay endoscopic removal of foreign body when necessary.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Adverse drug reactions reported for injection use unless otherwise noted. Adverse reactions reported in children, adolescents, and adults.

>10%:

Gastrointestinal: Nausea (intranasal, IV: 26% to 30%), vomiting (intranasal, IV: 15% to 16%)

Nervous system: Headache (intranasal: 18%; IV: 5%)

Respiratory: Upper respiratory system symptoms (intranasal: including cough, epistaxis, nasal congestion, nasal discomfort, sneezing, throat irritation: 12%)

1% to 10%:

Immunologic: Antibody development (2%)

Local: Pain at injection site (1%), swelling at injection site (7%)

Frequency not defined (any route of administration):

Cardiovascular: Hypertension, hypotension (up to 2 hours after GI procedures), tachycardia

Dermatologic: Pallor, pruritus (including ears), pruritus of nose, urticaria

Endocrine & metabolic: Hot flash, hyperglycemia

Gastrointestinal: Abdominal pain, diarrhea, dysgeusia

Hypersensitivity: Hypersensitivity reaction (including anaphylactic shock)

Local: Injection-site reaction (including discomfort at injection site, erythema at injection site)

Nervous system: Altered sense of smell (intranasal), asthenia, dizziness, drowsiness

Ophthalmic: Eye pruritus, eye redness, watery eyes

Respiratory: Rhinorrhea

Postmarketing (any indication): Dermatologic: Necrolytic migratory erythema

Contraindications

Known hypersensitivity to glucagon or any component of the formulation (injection contains lactose); pheochromocytoma; insulinoma; use of injection as a diagnostic aid in patients with glucagonoma.

Warnings/Precautions

Concerns related to adverse effects:

- Hypersensitivity reactions: Allergic reactions including skin rash and anaphylactic shock (with hypotension and respiratory difficulties) have been reported; reactions have generally been associated with endoscopic patients. Discontinue use and institute appropriate therapy in patients with severe hypersensitivity reactions.
- Necrolytic migratory erythema: Necrolytic migratory erythema (NME), a skin rash associated with glucagonomas (glucagon-producing tumors) and characterized by scaly, pruritic, erythematous plaques, bullae, and erosions, has been reported (rarely) following continuous glucagon infusion. Rash may occur on face, groin, perineum, and legs or may be more widespread; rash generally resolves with discontinuation of treatment. Consider the risks versus benefits of continuing the glucagon infusion if NME occurs.

Disease-related concerns:

- Adrenal insufficiency: Use with caution in patients with adrenal insufficiency; hepatic glycogen levels may be inadequate for glucagon to effectively increase blood glucose.
- Cardiac disease: Use with caution in patients with cardiac disease.
- Chronic hypoglycemia: Use with caution in patients with chronic hypoglycemia; hepatic glycogen levels may be inadequate for glucagon to effectively increase blood glucose.
- Diabetes: Use caution if using as diagnostic aid in patients with diabetes on insulin; may cause hyperglycemia.
- Glucagonoma: Use with caution in patients with glucagonoma; may cause secondary hypoglycemia. The use of injectable glucagon as a diagnostic aid is contraindicated in patients with this condition.
- Insulinoma: Exogenous glucagon may cause an initial rise in blood glucose followed by rebound hypoglycemia. The use of glucagon is contraindicated in patients with this condition.
- Pheochromocytoma: Exogenous glucagon may cause the release of catecholamines, resulting in an increase in blood pressure. The use of glucagon is contraindicated in patients with this condition.
- Starvation/fasting: Use caution with prolonged fasting and/or starvation; hepatic glycogen levels may be inadequate for glucagon to effectively increase blood glucose.

Dosage form specific issues:

- Lactose: May contain lactose; avoid administration in hereditary galactose intolerance, congenital lactase deficiency, or glucose-galactose malabsorption.

Other warnings/precautions:

- Appropriate use: Insulin or sulfonylurea overdose: Patients with hypoglycemia should immediately be treated with dextrose. If IV access cannot be established or if dextrose is not available, glucagon may be considered as alternative acute treatment until dextrose can be administered.
- Secondary hypoglycemia: Supplemental carbohydrates should be given to patients who respond to glucagon for severe hypoglycemia to prevent secondary hypoglycemia.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Powder, Nasal:

Baqsimi One Pack: 3 mg/dose (1 ea)

Baqsimi Two Pack: 3 mg/dose (1 ea)

Powder, Nasal [preservative free]:

Baqsimi One Pack: 3 mg/dose (1 ea [DSC])

Baqsimi Two Pack: 3 mg/dose (1 ea [DSC])

Solution, Intravenous:

Gvoke VialDx: 1 mg/0.2 mL (0.2 mL [DSC]) [contains dimethyl sulfoxide]

Solution, Subcutaneous [preservative free]:

Gvoke Kit: 1 mg/0.2 mL (0.2 mL)

Solution Auto-injector, Subcutaneous:

Gvoke HypoPen 1-Pack: 0.5 mg/0.1 mL (0.1 mL); 1 mg/0.2 mL (0.2 mL)

Gvoke HypoPen 2-Pack: 0.5 mg/0.1 mL (0.1 mL); 1 mg/0.2 mL (0.2 mL)

Solution Prefilled Syringe, Subcutaneous [preservative free]:

Gvoke PFS: 0.5 mg/0.1 mL (0.1 mL); 1 mg/0.2 mL (0.2 mL)

Solution Reconstituted, Injection:

GlucaGen Diagnostic: 1 mg (1 ea [DSC])

GlucaGen HypoKit: 1 mg (1 ea [DSC])

Generic: 1 mg (1 ea); 1 mg/mL (1 ea)

Solution Reconstituted, Injection [preservative free]:

Generic: 1 mg (1 ea)

Generic Equivalent Available: US

May be product dependent

Pricing: US

Powder (Baqsimi One Pack Nasal)

3 mg/dose (per each): \$368.18

Powder (Baqsimi Two Pack Nasal)

3 mg/dose (per each): \$368.19

Solution (Gvoke Kit Subcutaneous)

1 mg/0.2 mL (per 0.2 mL): \$402.34

Solution (reconstituted) (Glucagon HCl (Diagnostic) Injection)

1 mg (per each): \$180.00 - \$303.22

Solution Auto-injector (Gvoke HypoPen 1-Pack Subcutaneous)

0.5MG/0.1ML (per 0.1 mL): \$402.34

1 mg/0.2 mL (per 0.2 mL): \$402.34

Solution Auto-injector (Gvoke HypoPen 2-Pack Subcutaneous)

0.5MG/0.1ML (per 0.1 mL): \$402.34

1 mg/0.2 mL (per 0.2 mL): \$402.34

Solution Prefilled Syringe (Gvoke PFS Subcutaneous)

1 mg/0.2 mL (per 0.2 mL): \$402.34

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Powder, Nasal:

Baqsimi: 3 mg/dose (1 ea)

Administration: Adult

Note: For the treatment of hypoglycemia, administer fast-acting and long-acting oral carbohydrates to patient as soon as possible after response to glucagon treatment. If patient is unconscious, place in lateral recumbent position to prevent choking when consciousness returns.

IM: For treatment of hypoglycemia, may administer reconstituted solution for injection as an IM injection in the upper arms, thighs, or buttocks. For use as a diagnostic aid for radiologic examinations, may administer IM; for examination of the colon, administer ~10 minutes prior to procedure. After the diagnostic procedure, administer oral carbohydrates to patients who have been fasting, if this is compatible with the diagnostic procedure applied.

Intranasal: Insert the tip of the intranasal device into one nostril and fully depress plunger until the green line is no longer visible. Inhalation of the dose is not necessary. Do not reuse the device after administration.

IV: Rapid injection may be associated with increased nausea and vomiting.

Anaphylaxis (refractory to epinephrine) in patients on beta-blocker therapy: Administer IV bolus over 5 minutes; continuous infusions may be used (Ref).

Beta-blocker/calcium channel blocker toxicity: Administer IV bolus over 3 to 5 minutes; continuous infusions may be used. Ensure adequate supply available to continue therapy (Ref).

Diagnostic aid for radiologic examinations: Administer IV push over 1 minute. After the diagnostic procedure, administer oral carbohydrates to patients who have been fasting, if this is compatible with the diagnostic procedure applied.

Hypoglycemia: May administer IV.

SUBQ:

Injection, powder for reconstitution: For treatment of hypoglycemia, may administer reconstituted solution in the upper arms, thighs, or buttocks.

Injection, solution (Gvoke HypoPen, Gvoke PFS, Gvoke Kit): Administer in the lower abdomen, outer thigh, or outer upper arm; inject rapidly to reduce injection-site pain.

Preparation for Administration: Adult

Injection, powder for reconstitution: Reconstitute powder for injection by adding 1 mL of manufacturer-supplied sterile diluent or sterile water for injection to a vial containing 1 mg of the drug, to provide solutions containing 1 mg/mL. Shake vial gently to dissolve. A solution for continuous infusion may be prepared by further dilution in D5W (Ref); also refer to institution-specific policies and procedures.

Injection, solution (Gvoke VialDx): Dilute with NS prior to administration. Withdraw 0.2 mL from vial and add to separate sterile container containing 2 mL of NS to make a diluted solution with a final concentration of 0.45 mg/mL. Gently swirl the container to thoroughly mix solution. Not approved for continuous IV infusion.

Usual Infusion Concentrations: Adult

IV infusion: 4 mg in 50 mL (concentration: 0.08 mg/mL); some centers may use a higher concentration (eg, 20 mg in 100 mL [concentration 0.2 mg/mL] of D5W) to meet the needs in some patients (eg, beta-blocker or calcium channel blocker overdose).

Administration: Pediatric

Hypoglycemia: Note: Administer fast-acting and long-acting oral carbohydrates to patient as soon as possible after response to treatment. If patient is unconscious, place patient on side to prevent choking if vomiting occurs.

Parenteral:

Vial: May administer IV or as IM or SUBQ injection in the upper arms, thighs, or buttocks.

Mini-dose regimen: Following reconstitution of 1 mg/mL solution, inject glucagon SUBQ using an insulin syringe (U-100); 1 unit on the syringe provides 0.01 mg of glucagon (Ref).

Autoinjector/prefilled syringe (Gvoke): For SUBQ use only. Do not open foil package until ready to administer. Administer SUBQ in the lower abdomen, outer thigh, or outer upper arm. Do not reuse the device after administration; a new device should be used for each dose.

Intranasal: For intranasal use only. Insert the tip of the device into 1 nostril and fully depress plunger until the green line is no longer visible. Inhalation of the dose is not necessary. Do not reuse the device after administration.

Anaphylaxis: Vial: IV: Initial dose should be administered over 5 minutes (Ref).

Beta-blocker/calcium channel blocker toxicity: Vial: IV: Initial loading dose (bolus) should be administered over 1 to 5 minutes (Ref); may also administer as a continuous IV infusion; ensure adequate supply available to continue therapy (Ref).

Preparation for Administration: Pediatric

Parenteral: Vial: Reconstitute powder for injection by adding 1 mL of manufacturer-supplied sterile diluent (diluent may vary by product; see product-specific labeling for details) or SWFI to the vial containing 1 mg of the drug; shake vial gently to dissolve powder; resultant concentration is 1 mg/mL. A solution for continuous IV infusion may be prepared by further diluting glucagon in D5W (Ref).

Storage/Stability

Injection, powder for reconstitution: Prior to reconstitution, store at 20°C to 25°C (69°F to 77°F) for up to 24 months. Do not freeze. Protect from light. Use reconstituted solution immediately; discard unused portion.

Injection, solution (Gvoke): Store at 20°C to 25°C (68°F to 77°F); excursions permitted between 15°C and 30°C (59°F and 86°F). Do not refrigerate, freeze, or expose to extreme temperatures. Store the specific products in the original containers as follows:

Gvoke HypoPen; Gvoke PFS (prefilled syringes): Store in the original sealed foil pouch until time of use.

Gvoke Kit: Store the vial and pouched syringe together in the original carton until time of use.

Gvoke VialDx: Store vials in the original carton until time of use.

Intranasal: Store at ≤30°C (86°F). Do not remove intranasal device from the provided shrink-wrapped tube until ready to use. For single use only; discard after use.

Pharmacy supply of emergency antidotes: Guidelines suggest that at least 90 to 250 mg of glucagon be stocked. Suggested amount is stated to be a sufficient quantity to treat 1 patient weighing 100 kg for an initial 8- to 24-hour period (Dart 2018); actual amount to be stocked should take into account site-specific and population-specific needs.

Use: Labeled Indications

Diagnostic aid: Radiologic examinations (injection): As a diagnostic aid during radiologic examinations to temporarily inhibit movement of the GI tract in adults.

Hypoglycemia: Note: The American Diabetes Association (ADA) recommends that glucagon be prescribed for all patients with diabetes at increased risk of level 2 hypoglycemia (<54 mg/dL); caregivers, school personnel, or family members of these patients should be trained on when and how to administer glucagon (ADA 2023).

Injection, powder for reconstitution (GlucaGen HypoKit or Glucagon Emergency): Treatment of severe hypoglycemia in pediatric and adult patients with diabetes.

Limitations of use: Products not packaged with a syringe and diluent necessary for rapid preparation and administration during an emergency outside of a health care facility are not indicated for the emergency treatment of hypoglycemia.

Injection, solution (Gvoke HypoPen, Gvoke PFS, Gvoke Kit): Treatment of severe hypoglycemia in adult and pediatric patients (≥2 years of age) with diabetes.

Intranasal: Treatment of severe hypoglycemia in adult and pediatric patients (≥1 year of age) with diabetes.

Use: Off-Label: Adult

Anaphylaxis in patients on beta-blockers; Beta-blocker overdose; Calcium channel blocker overdose; Growth hormone deficiency (alternative diagnostic test)

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Agents with Clinically Relevant Anticholinergic Effects: May increase adverse/toxic effects of Glucagon. Specifically, the risk of gastrointestinal adverse effects may be increased. *Risk C: Monitor*

Antidiabetic Agents: Hyperglycemia-Associated Agents may decrease therapeutic effects of Antidiabetic Agents. *Risk C: Monitor*

Indomethacin (Systemic): May decrease therapeutic effects of Glucagon and Glucagon Analogs. *Risk C: Monitor*

Vitamin K Antagonists: Glucagon and Glucagon Analogs may increase anticoagulant effects of Vitamin K Antagonists. *Risk C: Monitor*

Food Interactions

Glucagon depletes glycogen stores.

Pregnancy Considerations

Glucagon may be used for the treatment of severe hypoglycemia during pregnancy (Alexopoulos 2019). In general, medications used as antidotes should take into consideration the health and prognosis of the mother; antidotes should be administered to pregnant females if there is a clear indication for use and should not be withheld because of concerns of teratogenicity (Bailey 2003a).

Breastfeeding Considerations

Glucagon is not absorbed from the GI tract and therefore, it is unlikely adverse effects would occur in a breastfeeding infant.

Dietary Considerations

Administer oral carbohydrates to patient as soon as possible after response to treatment.

Monitoring Parameters

BP, blood glucose, ECG, heart rate, mentation; signs or symptoms of a hypersensitivity reaction.

Growth hormone (GH) deficiency: Serum GH levels at baseline (prior to glucagon injection) and every 30 minutes for 4 hours following administration (AACE/ACE [Yuen 2019]; Hamrahian 2016).

Reference Range

Classification of hypoglycemia (ADA 2023):

Level 1: Glucose ≥ 54 to ≤ 70 mg/dL (SI: ≥ 3 to ≤ 3.9 mmol/L); hypoglycemia alert value; initiate fast-acting carbohydrate (eg, glucose) treatment.

Level 2: Glucose < 54 mg/dL (SI: < 3 mmol/L); threshold for neuroglycopenic symptoms; requires immediate action.

Level 3: Hypoglycemia associated with a severe event characterized by altered mental and/or physical status requiring assistance.

Diagnosis of adult growth hormone deficiency (AACE/ACE [Yuen 2019]): **Note:** Use caution when interpreting growth hormone (GH) levels in patients with hyperglycemia and in patients > 70 years of age; diagnostic accuracy of glucagon stimulation in these populations is unknown.

Patients with BMI < 25 kg/m², or with BMI 25 to 30 kg/m² and high pretest probability: Peak GH levels < 3 mcg/L (SI: < 3 ng/mL) (assay dependent) within 4 hours following glucagon administration confirm the presence of adult growth hormone deficiency.

Patients with BMI > 30 kg/m², or with BMI 25 to 30 kg/m² and low pretest probability: Peak GH levels < 1 mcg/L (SI: < 1 ng/mL) (assay dependent) within 4 hours following glucagon administration confirm the presence of adult growth hormone deficiency.

Mechanism of Action

Stimulates adenylate cyclase to produce increased cyclic AMP, which promotes hepatic glycogenolysis and gluconeogenesis, causing a raise in blood glucose levels; antihypoglycemic effect requires pre-existing hepatic glycogen stores. Extra hepatic effects of glucagon include relaxation of the smooth muscle of the stomach, duodenum, small bowel, and colon.

In the setting of beta-blocker and calcium channel blocker toxicity, the glucagon-mediated increase in cyclic AMP increases automaticity at the sinoatrial and atrioventricular nodes. In addition, glucagon improves myocardial contractility and produces peripheral vasodilation (Bailey 2003b).

Pharmacokinetics (Adult Data Unless Noted)

Onset of action:

Blood glucose concentrations: **Note:** In general, when used for the treatment of hypoglycemia, the time required to correct hypoglycemia (eg, increase blood glucose concentrations by 20 mg/dL) is similar between formulations and routes of administration (~10 to 15 minutes); clinicians should note that the durability of response may result in blood glucose concentrations that continue to rise for >90 minutes (Baqsimi prescribing information 2019; Gvoke prescribing information; Rickels 2016).

IV: Peak effect: 5 to 20 minutes (GlucaGen prescribing information 2018).

IM:

Onset: 10 minutes.

Peak effect: 30 minutes when used as a diagnostic aid (GlucaGen prescribing information 2018); in patients with type 1 diabetes and insulin-induced hypoglycemia, the peak glucose response was not yet reached during the 90-minute observation window following glucagon administration (Rickels 2016).

Intranasal:

Onset: Slightly delayed compared to IM administration; for example, median response time to reach glucose target in adults was 13 minutes (IM) versus 16 minutes (intranasal), but clinical outcomes/resolution of hypoglycemia at 30 minutes did not differ (Rickels 2016).

Peak effect: Patients with type 1 diabetes:

Children and Adolescents <17 years: ~60 minutes (Baqsimi prescribing information 2025)

Adults: A peak glucose response was not yet reached during the 90-minute observation window following glucagon administration (Baqsimi prescribing information 2019; Rickels 2016).

SUBQ:

Onset: ~10 minutes; Gvoke onset is delayed a few minutes compared to Glucagon Emergency, but clinical outcomes/resolution of hypoglycemia at 30 minutes did not differ between SUBQ formulations (Gvoke prescribing information).

Peak effect: 30 to 45 minutes in healthy patients (GlucaGen prescribing information 2018; Glucagon Emergency prescribing information 2018); in patients with type 1 diabetes and insulin-induced hypoglycemia, the peak glucose response was not yet reached during the 90-minute observation window following glucagon administration (Gvoke prescribing information).

GI relaxation: IV: 45 seconds; IM: 4 to 10 minutes.

Duration:

Blood glucose concentrations: 60 to 90 minutes (GlucaGen prescribing information 2018); other data suggest glucose elevation may persist >90 minutes following intranasal, IM, or SUBQ administration in patients with type 1 diabetes (Gvoke prescribing information; Rickels 2016).

GI relaxation: IV: 9 to 25 minutes; IM: 12 to 32 minutes.

Distribution: V_d : Reported values vary significantly: ~0.25 L/kg (Glucagon Emergency); 137 to 2,425 L (Gvoke); 885 L (Baqsimi).

Metabolism: Primarily hepatic; some inactivation occurring renally and in plasma.

Half-life elimination, plasma:

IV: 8 to 18 minutes.

IM (apparent): 26 to 45 minutes.

Intranasal (apparent): Median:

Children and Adolescents <17 years: 21 to 31 minutes.

Adults: ~35 minutes.

SUBQ: 32 minutes (Gvoke prescribing information; not reported for other formulations).

Time to peak, plasma:

IM: ~10 to 12.5 minutes.

Intranasal:

Children and Adolescents <17 years: 10 to 20 minutes.

Adults: 15 minutes.

SUBQ:

Glucagon Emergency: 20 minutes.

Gvoke:

Pediatric patients:

2 to <6 years: 41 minutes.

6 to <12 years: 34 minutes.

12 to <18 years: 51 minutes.

Adults: 50 minutes.

Patient Counseling Points

(For additional information see "Glucagon: Patient drug information")

What is this drug used for?

- It is used to treat low blood sugar.
- It is used to quiet the GI (gastrointestinal) tract so it may be x-rayed.
- It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

Injection (Glucagen):

- Irritation where the shot is given
- Feeling tired or weak
- Pale skin
- Diarrhea, upset stomach or throwing up
- Headache

Injection (Gvoke VialDx):

- Upset stomach

- Change in taste
- Dizziness
- Flushing
- Headache

Injection (Gvoke):

- Irritation where the shot is given
- Itching
- Upset stomach or throwing up
- Stomach pain
- Headache

Nose spray:

- Nose or throat irritation
- Runny or stuffy nose
- Cough
- Nosebleed
- Sneezing
- Red, watery, or itchy eyes
- Upset stomach or throwing up
- Headache

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

For all uses of this drug:

- Low blood sugar like dizziness, headache, feeling sleepy, feeling weak, shaking, a fast heartbeat, confusion, hunger, or sweating
- High or low blood pressure like very bad headache or dizziness, passing out, or change in eyesight
- A fast heartbeat
- Blisters; scaly, red, itchy, or painful skin; or if the skin starts to break down
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

For a procedure:

- High blood sugar like confusion, feeling sleepy, unusual thirst or hunger, passing urine more often, flushing, fast breathing, or breath that smells like fruit

For treating low blood sugar in children:

- High blood sugar like confusion, feeling sleepy, unusual thirst or hunger, passing urine more often, flushing, fast breathing, or breath that smells like fruit

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the

knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AT) Austria: Ogluo;
- (BR) Brazil: Glucagon;
- (CH) Switzerland: Glucagen;
- (CN) China: Glucagen;
- (DE) Germany: Ogluo;
- (FI) Finland: Glucagen | Glucagon lilly;
- (FR) France: Glucagon Dci;
- (GB) United Kingdom: Glucagon lilly | Ogluo;
- (HK) Hong Kong: Glucagon novo nordisk;
- (HU) Hungary: Glucagon;
- (IE) Ireland: Ogluo;
- (IN) India: Glucagon;
- (IT) Italy: Glucagen;
- (JO) Jordan: Glucagon;
- (KR) Korea, Republic of: Glucagon;
- (KW) Kuwait: Glucagen;
- (MY) Malaysia: Glucagon;
- (NL) Netherlands: Glucagen | Glucagon lilly | Ogluo;
- (NO) Norway: Glucagon | Ogluo;
- (PR) Puerto Rico: Baqsimi | Gvoke hypopen | Gvoke kit | Gvoke pfs;
- (PY) Paraguay: Glucagen;
- (QA) Qatar: Baqsimi | GlucaGen | GlucaGen Hypokit;
- (SA) Saudi Arabia: Glucagon;
- (SE) Sweden: Glucagon novo nordisk;
- (SG) Singapore: Glucagon;
- (TH) Thailand: Glucagon;
- (TW) Taiwan: Glucagon;
- (ZA) South Africa: Glucagen

Use of UpToDate is subject to the Terms of Use.

REFERENCES

1. Abraham MB, Karges B, Dovc K, et al. ISPAD clinical practice consensus guidelines 2022: assessment and management of hypoglycemia in children and adolescents with diabetes. *Pediatr Diabetes*. 2022;23(8):1322-1340. doi:10.1111/pedi.13443 [PubMed 36537534]
2. Alexopoulos AS, Blair R, Peters AL. Management of preexisting diabetes in pregnancy: a review. *JAMA*. 2019;321(18):1811-1819. doi:10.1001/jama.2019.4981 [PubMed 31087027]

3. American Diabetes Association (ADA). Standards of care in diabetes-2023. *Diabetes Care*. 2023;46(suppl 1):S1-S291. https://diabetesjournals.org/care/issue/46/Supplement_1. Accessed January 4, 2023.
4. American Diabetes Association (ADA). Standards of care in diabetes-2025. *Diabetes Care*. 2025;48(suppl 1):S1-S336. https://diabetesjournals.org/care/issue/48/Supplement_1. Accessed May 8, 2025.
5. Arroyo AM, Kao LW. Calcium channel blocker toxicity. *Pediatr Emerg Care*. 2009;25(8):532-540. [PubMed 19687715]
6. ASGE Standards of Practice Committee, Ikenberry SO, Jue TL, et al. Management of ingested foreign bodies and food impactions. *Gastrointest Endosc*. 2011;73(6):1085-1091. [PubMed 21628009]
7. Aynsley-Green A, Hussain K, Hall J, et al. Practical Management of Hyperinsulinism in Infancy. *Arch Dis Child Fetal Neonatal Ed*. 2000;82(2):98-107. [PubMed 10685981]
8. Bailey B. Are there teratogenic risks associated with antidotes used in the acute management of poisoned pregnant women? *Birth Defects Res A Clin Mol Teratol*. 2003a;67(2):133-140. doi:10.1002/bdra.10007 [PubMed 12769509]
9. Bailey B. Glucagon in beta-blocker and calcium channel blocker overdoses: a systematic review. *J Toxicol Clin Toxicol*. 2003b;41(5):595-602. doi:10.1081/clt-120023761 [PubMed 14514004]
10. Baqsimi (glucagon) [prescribing information]. Rancho Cucamonga, CA: Amphastar Pharmaceuticals; April 2025.
11. Barrueto F. Beta blocker poisoning. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed June 12, 2024a.
12. Barrueto F. Calcium channel blocker poisoning. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed June 12, 2024b.
13. Campbell RL, Li JT, Nicklas RA, Sadosty AT; Members of the Joint Task Force; Practice Parameter Workgroup. Emergency department diagnosis and treatment of anaphylaxis: a practice parameter. *Ann Allergy Asthma Immunol*. 2014;113(6):599-608. [PubMed 25466802]
14. Cardona V, Ansotegui IJ, Ebisawa M, et al. World Allergy Organization anaphylaxis guidance 2020. *World Allergy Organ J*. 2020;13(10):100472. doi:10.1016/j.waojou.2020.100472 [PubMed 33204386]
15. Carter PE, Lloyd DJ, Duffty P. Glucagon for hypoglycaemia in infants small for gestational age. *Arch Dis Child*. 1988;63(10):1264-1266. [PubMed 3196055]
16. Charsha DS, McKinley PS, Whitfield JM. Glucagon infusion for treatment of hypoglycemia: efficacy and safety in sick, preterm infants. *Pediatrics*. 2003;111(1):220-221. [PubMed 12509583]
17. Cheng A. Emergency treatment of anaphylaxis in infants and children. *Paediatr Child Health*. 2011;16(1):35-40. [PubMed 22211074]
18. Chiang JL, Maahs DM, Garvey KC, et al. Type 1 diabetes in children and adolescents: a position statement by the American Diabetes Association. *Diabetes Care*. 2018;41(9):2026-2044. [PubMed 30093549]
19. Chung ST, Haymond MW. Minimizing morbidity of hypoglycemia in diabetes: a review of mini-dose glucagon. *J Diabetes Sci Technol*. 2015;9(1):44-51. [PubMed 25160667]
20. Dart RC, Goldfrank LR, Erstad BL, et al. Expert consensus guidelines for stocking of antidotes in hospitals that provide emergency care. *Ann Emerg Med*. 2018;71(3):314-325.e1. doi:10.1016/j.annemergmed.2017.05.021 [PubMed 28669553]
21. Dichtel LE, Yuen KC, Bredella MA, et al. Overweight/Obese adults with pituitary disorders require lower peak growth hormone cutoff values on glucagon stimulation testing to avoid over-

- diagnosis of growth hormone deficiency. *J Clin Endocrinol Metab.* 2014;99(12):4712-4719. doi:10.1210/jc.2014-2830 [PubMed 25210883]
22. Dirir H, Karaca Z, Simsek Y, et al. Can a glucagon stimulation test characterized by lower GH cut-off value be used for the diagnosis of growth hormone deficiency in adults? *Pituitary.* 2015;18(6):884-892. doi:10.1007/s11102-015-0666-1 [PubMed 26129876]
 23. Field JM, Hazinski MF, Sayre MR, et al. Part 1: executive summary: 2010 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. *Circulation.* 2010;122(18)(suppl 3):S640-S656. [PubMed 20956217]
 24. Fung BM, Sweetser S, Wong Kee Song LM, Tabibian JH. Foreign object ingestion and esophageal food impaction: An update and review on endoscopic management. *World J Gastrointest Endosc.* 2019;11(3):174-192. [PubMed 30918584]
 25. Galcheva S, Al-Khawaga S, Hussain K. Diagnosis and management of hyperinsulinaemic hypoglycaemia. *Best Pract Res Clin Endocrinol Metab.* 2018;32(4):551-573. [PubMed 30086874]
 26. GlucaGen (glucagon) [prescribing information]. Ridgefield, CT: Boehringer Ingelheim Pharmaceuticals Inc; July 2021.
 27. GlucaGen HypoKit and Diagnostic Kit (glucagon) [prescribing information]. Plainsboro, NJ: Novo Nordisk; March 2021.
 28. Glucagon Emergency Kit [prescribing information]. Lake Zurich, IL: Fresenius Kabi; August 2020.
 29. Glucagon for injection [prescribing information]. Lake Zurich, IL: Fresenius Kabi; July 2022.
 30. Glucagon injection [prescribing information]. Lake Zurich, IL: Fresenius Kabi; January 2016.
 31. Glucagon injection [prescribing information]. Lake Zurich, IL: Fresenius Kabi; August 2020.
 32. Golden DBK, Wang J, Wasserman S, et al; Joint Task Force on Practice Parameters Reviewers; Abrams EM, Bernstein JA, Chu DK, et al. Anaphylaxis: a 2023 practice parameter update. *Ann Allergy Asthma Immunol.* 2024;132(2):124-176. doi:10.1016/j.anai.2023.09.015 [PubMed 38108678]
 33. Grøndahl MF, Lange AH, Suppli MP, et al. Glucagon clearance is decreased in chronic kidney disease but preserved in liver cirrhosis. *Diabetes.* 2024;73(10):1641-1647. doi:10.2337/db24-0305 [PubMed 39052774]
 34. Gvoke (glucagon) [prescribing information]. Chicago, IL: Xeris Pharmaceuticals Inc; December 2023.
 35. Gvoke and Gvoke VialDx (glucagon) [prescribing information]. Shirley, NY: American Regent Inc; March 2025.
 36. Hamrahian AH, Yuen KC, Gordon MB, Pulaski-Liebert KJ, Bena J, Biller BM. Revised GH and cortisol cut-points for the glucagon stimulation test in the evaluation of GH and hypothalamic-pituitary-adrenal axes in adults: results from a prospective randomized multicenter study. *Pituitary.* 2016;19(3):332-341. doi:10.1007/s11102-016-0712-7 [PubMed 26897383]
 37. Hawdon JM, Aynsley-Green A, Ward Platt MP. Neonatal Blood Glucose Concentrations: Metabolic Effects of Intravenous Glucagon and Intragastric Medium Chain Triglyceride. *Arch Dis Child.* 1993;68(3 Spec No):255-261. [PubMed 8466259]
 38. Haymond MW, Schreiner B. Mini-dose glucagon rescue for hypoglycemia in children with type 1 diabetes. *Diabetes Care.* 2001;24(4):643-645. [PubMed 11315823]
 39. Holger JS, Engebretsen KM, Obetz CL, et al. A Comparison of Vasopressin and Glucagon in Beta-Blocker Induced Toxicity. *Clin Toxicol (Phila).* 2006;44(1):45-51. doi:10.1080/15563650500394795 [PubMed 16496493]
 40. Hussain K, Aynsley-Green A. Hyperinsulinaemic Hypoglycaemia in Preterm Neonates. *Arch Dis Child Fetal Neonatal Ed.* 2004;89(1):65-67. [PubMed 14711860]

41. Kapoor RR, Flanagan SE, James C, et al. Hyperinsulinaemic Hypoglycaemia. *Arch Dis Child*. 2009;94(6):450-457. [PubMed 19193661]
42. Kleinman ME, Chameides L, Schexnayder SM, et al. Part 14: pediatric advanced life support: 2010 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. *Circulation*. 2010;122(18)(suppl 3):S876-S908. [PubMed 20956230]
43. Kusumoto FM, Schoenfeld MH, Barrett C, et al. 2018 ACC/AHA/HRS Guideline on the Evaluation and Management of Patients With Bradycardia and Cardiac Conduction Delay: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines and the Heart Rhythm Society. *J Am Coll Cardiol*. 2019;74(7):e51-e156. doi:10.1016/j.jacc.2018.10.044 [PubMed 30412709]
44. Lavonas EJ, Akpunonu PD, Arens AM, et al; American Heart Association. 2023 American Heart Association focused update on the management of patients with cardiac arrest or life-threatening toxicity due to poisoning: an update to the American Heart Association guidelines for cardiopulmonary resuscitation and emergency cardiovascular care. *Circulation*. 2023;148(16):e149-e184. doi:10.1161/CIR.0000000000001161 [PubMed 37721023]
45. Lieberman P, Nicklas RA, Randolph C, et al. Anaphylaxis--a practice parameter update 2015. *Ann Allergy Asthma Immunol*. 2015;115(5):341-384. doi:10.1016/j.anai.2015.07.019 [PubMed 26505932]
46. Love JN, Sachdeva DK, Bessman ES, et al. A Potential Role for Glucagon in the Treatment of Drug-Induced Symptomatic Bradycardia. *Chest*. 1998;114(1):323-326. [PubMed 9674488]
47. Mehta A, Wootton R, Cheng KN, et al. Effect of Diazoxide or Glucagon on Hepatic Glucose Production Rate During Extreme Neonatal Hypoglycaemia. *Arch Dis Child*. 1987;62(9):924-930. [PubMed 3314727]
48. Miralles RE, Lodha A, Perlman M, et al. Experience With Intravenous Glucagon Infusions as a Treatment for Resistant Neonatal Hypoglycemia. *Arch Pediatr Adolesc Med*. 2002;156(10):999-1004. [PubMed 12361445]
49. Mokhlesi B, Leikin JB, Murray P, et al. Adult Toxicology in Critical Care: Part II: Specific Poisonings. *Chest*. 2003;123(3):897-922. [PubMed 12628894]
50. Nelson LS, Howland, MA, Lewin NA, et al, eds. *Goldfrank's Toxicologic Emergencies*. 11th ed. McGraw-Hill Education; 2019.
51. Panchal AR, Bartos JA, Cabañas JG, et al; Adult Basic and Advanced Life Support Writing Group. Part 3: adult basic and advanced life support: 2020 American Heart Association guidelines for cardiopulmonary resuscitation and emergency cardiovascular care. *Circulation*. 2020;142(16)(suppl 2):S366-S468. doi:10.1161/CIR.0000000000000916 [PubMed 33081529]
52. Phelan H, Hanas R, Hofer SE, et al. Sick day management in children and adolescents with diabetes. *Pediatr Diabetes*. 2022;23(7):912-925. doi:10.1111/pedi.13415 [PubMed 36093857]
53. Refer to manufacturer's labeling.
54. Rickels MR, Ruedy KJ, Foster NC, et al; T1D Exchange Intranasal Glucagon Investigators. Intranasal glucagon for treatment of insulin-induced hypoglycemia in adults with type 1 diabetes: a randomized crossover noninferiority study. *Diabetes Care*. 2016;39(2):264-270. doi: 10.2337/dc15-1498. [PubMed 26681725]
55. Shah P, Rahman SA, Demirbilek H, Güemes M, Hussain K. Hyperinsulinaemic hypoglycaemia in children and adults. *Lancet Diabetes Endocrinol*. 2017;5(9):729-742. [PubMed 27915035]
56. Shaker MS, Wallace DV, Golden DBK, et al. Anaphylaxis-a 2020 practice parameter update, systematic review, and Grading of Recommendations, Assessment, Development and Evaluation (GRADE) analysis. *J Allergy Clin Immunol*. 2020;145(4):1082-1123. doi:10.1016/j.jaci.2020.01.017 [PubMed 32001253]

57. Shenoi RP, Timm N; Committee on Drugs; Committee on Pediatric Emergency Medicine. Drugs Used to Treat Pediatric Emergencies. *Pediatrics*. 2020;145(1):e20193450. [PubMed 31871244]
58. Shepherd G. Treatment of poisoning caused by beta-adrenergic and calcium-channel blockers [published correction appears in *J Health Syst Pharm*. 2008;65(17):1592]. *Am J Health Syst Pharm*. 2006;63(19):1828-1835. doi:10.2146/ajhp060041 [PubMed 16990629]
59. St-Onge M, Anseeuw K, Cantrell FL, et al. Experts consensus recommendations for the management of calcium channel blocker poisoning in adults. *Crit Care Med*. 2017;45(3):e306-e315. doi:10.1097/CCM.0000000000002087 [PubMed 27749343]
60. Thornton PS, Stanley CA, De Leon DD, et al. Recommendations from the Pediatric Endocrine Society for Evaluation and Management of Persistent Hypoglycemia in Neonates, Infants, and Children. *J Pediatr*. 2015;167(2):238-245. [PubMed 25957977]
61. Trujillo MH, Guerrero J, Fragachan C, et al. Pharmacologic Antidotes in Critical Care Medicine: A Practical Guide for Drug Administration. *Crit Care Med*. 1998;26(2):377-391. doi:10.1097/00003246-199802000-00044 [PubMed 9468179]
62. Vanden Hoek TL, Morrison LJ, Shuster M, et al. Part 12: cardiac arrest in special situations: 2010 American Heart Association guidelines for cardiopulmonary resuscitation and emergency cardiovascular care [published corrections appear in *Circulation*. 2011;123(6):e239; *Circulation*. 2011;124(15):e405]. *Circulation*. 2010;122(18)(suppl 3):S829-S861. doi:10.1161/CIRCULATIONAHA.110.971069 [PubMed 20956228]
63. Wax PM, Erdman AR, Chyka PA, et al. Beta-Blocker Ingestion: An Evidence-Based Consensus Guideline for Out-of-Hospital Management. *Clin Toxicol (Phila)*. 2005;43(3):131-146. [PubMed 15906457]
64. Yorifuji T, Horikawa R, Hasegawa T, et al. Clinical practice guidelines for congenital hyperinsulinism. *Clin Pediatr Endocrinol*. 2017;26(3):127-152. [PubMed 28804205]
65. Yuen KC, Biller BM, Katznelson L, et al. Clinical characteristics, timing of peak responses and safety aspects of two dosing regimens of the glucagon stimulation test in evaluating growth hormone and cortisol secretion in adults. *Pituitary*. 2013;16(2):220-230. doi:10.1007/s11102-012-0407-7 [PubMed 22806554]
66. Yuen KCJ, Biller BMK, Radovick S, et al. American Association Of Clinical Endocrinologists and American College of Endocrinology guidelines for management of growth hormone deficiency in adults and patients transitioning from pediatric to adult care. *Endocr Pract*. 2019;25(11):1191-1232. doi:10.4158/GL-2019-0405 [PubMed 31760824]

Topic 8493 Version 413.0

Source: UpToDate. Downloaded 2026-05-26.